



Maternal insulin resistance during late gestation impairs brown adipose tissue development and thermogenic function in calves

Rui He,^{1,2*} Guiling Ma,^{3*} Zhiyong Hu,^{4*} Yang Gai,^{1,2} Yu Zhang,^{1,2} Shilong Huang,^{1,2} Shengyong Mao,^{1,2} Morteza Hosseini Ghaffari,^{5†} and Yanting Chen^{1,2†}

¹College of Animal Science and Technology, Nanjing Agricultural University, Nanjing 210095, China

²National Center for International Research on Animal Gut Nutrition, Jiangsu Key Laboratory of Gastrointestinal Nutrition and Animal Health, Nanjing Agricultural University, Nanjing, 210095 China

³College of Agro-Grassland Science, Nanjing Agricultural University, Nanjing 210095, China

⁴College of Animal Science, Shandong Agricultural University, Taian 21018, China

⁵Research Institute for Farm Animal Biology (FBN), Dummerstorf 18196, Germany

ABSTRACT

This study investigated the effect of maternal insulin resistance (IR) during late gestation on brown adipose tissue (BAT) development and thermogenic function in neonatal calves. Forty-four dry Holstein cows were selected for the study. Blood samples were collected during the dry period to assess insulin and glucose concentrations, and maternal IR was evaluated using the homeostasis model assessment (HOMA-IR). The cows were divided into 2 groups: maternal low IR ($n = 22$) and maternal high IR ($n = 22$). Immediately after birth, calves were separated from their dams and closely monitored for growth. Calf health was monitored from birth to 1 mo of age, 3 times per week, using standardized clinical scoring of fecal consistency, cough, nasal discharge, and respiration rate. Body weight, body length, withers height, and rectal temperature were recorded at birth, and blood glucose levels were assessed at various intervals following a glucose tolerance test (GTT). At birth ($n = 10$ per group) and at 1 mo ($n = 12$ per group), calves were killed, and perirenal BAT samples were collected for histological and molecular analyses. Thermogenic and mitochondrial dynamics markers were evaluated at the mRNA and protein levels using quantitative PCR and western blotting. Stromal vascular fractions (SVF) from perirenal BAT collected from calves at 0 d of age were cultured under hyperglycemic and hyperinsulinemic conditions to assess their differentiation into brown adipocytes, lipid droplet formation, and gene expression of key thermogenic markers, including uncoupling protein

1 (UCP1) and dynamin-related protein 1 (DRP1). The results showed that maternal IR significantly increased serum insulin and glucose levels in cows but did not affect BCS or backfat thickness (BFT). At birth, calves from high-IR dams had significantly lower perirenal BAT mass and larger brown adipocytes. Gene expression of thermogenic markers, such as UCP1 and PGC-1 α , was downregulated, and mitochondrial density in BAT was significantly reduced in maternal-high-IR calves. These effects persisted at one month, with continued reductions in BAT mass, mitochondrial density, and thermogenic gene expression. Maternal-high-IR calves exhibited impaired glucose tolerance, with delayed glucose clearance in the GTT. Maternal-high-IR calves also showed increased clinical morbidity, including higher fecal scores, increased cough scores, and a greater incidence of diarrhea. In vitro, SVF from calves demonstrated significantly reduced lipid droplet formation and impaired thermogenic gene expression. Mitochondrial fission markers DRP1 and FIS1 were downregulated in the maternal-high-IR group. These results suggest that maternal IR during late gestation impairs BAT development, mitochondrial dynamics, and glucose metabolism in neonatal calves, leading to long-term consequences for thermogenic function and metabolic health.

Key words: calf, insulin resistance, brown adipose tissue, mitochondrial dynamics

INTRODUCTION

During late gestation, a rapid increase in hormone production induces maternal insulin resistance (IR), which significantly elevates both fasting and postprandial glucose levels in the dam (Bell and Bauman, 1997; De Koster and Opsomer, 2013). In dairy cows, this hyperglycemic state enhances transplacental glucose transfer to the fetus, potentially leading to fatal hyperinsulinemia (Metzger

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*These authors contributed equally to this work.

†Corresponding authors: hosseini-ghaffari@fbn-dummerstorf.de and yanting.chen@njau.edu.cn

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

et al., 2009; Micke et al., 2010). Accompanying IR, impaired maternal satiety signaling during late gestation may result in excessive energy intake, increasing circulating free fatty acids (FFA) that can cross the placenta and promote fetal fat accumulation (Hasegawa et al., 2019). Evidence from human and rodent models indicates that maternal hyperglycemia and IR contribute to fetal over nutrition and adiposity, thereby increasing the risk of fetal obesity (Kim et al., 2012). Fetal hyperinsulinemia and excess fat accumulation have been associated with long-term metabolic disorders, including obesity, type 2 diabetes mellitus, cardiovascular disease, and dysfunction of brown adipose tissue (BAT; Zhang et al., 2016; Helle et al., 2018; Esser et al., 2020). Despite this growing concern, few studies have systematically investigated the consequences of maternal hyperglycemia during late gestation on fetal development in bovine species.

Brown adipose tissue develops predominantly during late gestation and plays a critical role in thermogenesis and energy homeostasis in neonates (Reid et al., 2019; Maliszewska and Kretowski, 2021). Functionally, BAT facilitates the oxidation of glucose and FFA to generate heat, helping prevent neonatal hyperglycemia and hypothermia (Calderon-Dominguez et al., 2016). Beyond its metabolic function, BAT enables newborns to maintain body temperature through nonshivering thermogenesis, a process largely mediated by the high expression of uncoupling protein 1 (UCP1; Chouchani et al., 2019). Uncoupling protein 1 uncouples oxidative phosphorylation from ATP production, dissipating energy as heat, which increases metabolic rate and reduces the incidence of hypothermia in newborns (Symonds et al., 2011; Plush et al., 2016). However, the effect of maternal IR and hyperglycemia during late gestation on neonatal and postnatal BAT development and function in calves remains poorly understood.

The function of BAT is critically dependent on mitochondrial dynamics, including the process of mitochondrial fusion and fission (Michurina et al., 2021). Mitochondrial fusion, primarily regulated by mitofusins (MFN1/2), plays a vital role in maintaining UCP1-mediated thermogenesis and energy balance in BAT, particularly during the neonatal period (Boutant et al., 2017). Conversely, mitochondrial fission, largely controlled by dynamin-related protein 1 (DRP1) and fission protein 1 (FIS1), contributes to mitochondrial turnover, biogenesis, and the removal of damaged mitochondrial components under normal physiological conditions (Mooli et al., 2020; Ibayashi et al., 2024). Although a balance between fusion and fission is necessary for optimal BAT function, stressors such as hyperglycemia may disrupt this equilibrium, leading to impaired mitochondrial integrity and BAT dysfunction (Qiang et al., 2016; Lee et al., 2018; Viana-Huete et al., 2018). However, the specific effects of maternal hyperglycemia during late

gestation on postnatal mitochondrial dynamics in bovine BAT remain to be elucidated.

Although maternal IR and hyperglycemia have been linked to altered fetal growth and adiposity in several species (Kawashima et al., 2016; Xuan Nguyen et al., 2023), limited information is available regarding their effects on BAT mass, development, mitochondrial characteristics, and glucose metabolism in neonatal calves. Understanding how maternal metabolic status shapes early-life BAT function may provide insight into mechanisms underlying impaired thermogenic capacity and metabolic adaptability in dairy calves. Therefore, we hypothesized that maternal IR during late gestation alters BAT development and metabolic function in neonatal calves. The objective of this study was to investigate the effects of maternal IR during late gestation on the BAT mass and development, mitochondrial dynamics, and glucose metabolism in neonatal calves, and to determine how these alterations may contribute to long-term consequences for thermogenic function and metabolic health.

MATERIALS AND METHODS

This study was conducted at Shandong High Speed Modern Dairy Farm, located in Shandong Province, China, from September to December 2023. All animal procedures followed the Guide for the Care and Use of Laboratory Animals, and the study protocol was approved by the Institutional Animal Care and Use Committee (IACUC) of Nanjing Agricultural University (Protocol #NJAULLSC2022006). All experimental procedures complied with ethical guidelines for animal research established by the IACUC, including the killing of calves and collection of tissue samples following their death, which were performed with due consideration for animal welfare. Throughout the study, investigators remained blinded to group assignments during experimental procedures and statistical analysis.

Animals, Experimental Design, Dietary Chemical Analyses, and Housing and Feeding Conditions

A total of 44 dry cows were selected for this study, with cows excluded for any history of mastitis, lameness, or other diseases recorded during lactation, as these conditions could interfere with metabolic processes and affect study outcomes. These cows had similar parity (mean $\pm$ SEM; 1.9 ± 0.14) and BW (742 ± 7.77 kg). The mean dry period length was 66.80 ± 1.84 d and did not differ between treatment groups (maternal low IR, 66.47 ± 0.74 d vs. maternal high IR 67.14 ± 0.94 d, $P = 0.24$).

Following dry-off, cows were monitored throughout the entire dry period and housed in freestall barns with dry manure bedding, ensuring a clean, comfortable, and stress-

free environment. The cows were fed a TMR ad libitum, offered twice daily at 0800 and 1700 h, formulated to meet the nutrient requirements for dry cows as outlined by the *Nutrient Requirements of Dairy Cattle* (NRC, 2001).

Weekly TMR samples were collected during the dry period for chemical composition analysis (Supplemental Table S1, see Notes) following the procedures described by Van Soest et al. (1991). Dry matter intake was measured twice per week under fastened feeding conditions. Body weight was recorded weekly with a weigh bridge. Body condition score was assessed weekly using a 5-point scale by 2 independent trained technicians (Edmonson et al., 1989). Backfat thickness (BFT) was measured weekly using a portable B-ultrasound (IMV Imaging), according to the manufacturer's instructions.

To ensure homogeneity within the experimental group, fetal sex was confirmed as male via B-mode ultrasonography (Mindray DP-3300, Shenzhen Mindray Bio-Medical Electronics Co. Ltd.), with only male calves included in the study to minimize variability in offspring outcomes. Blood samples were collected at wk 1, 3, 5, and 7 of the dry period to assess metabolic dynamics. Serum insulin concentrations were measured using ELISA kits (#MM-5100301, Meimian), and blood glucose concentrations were measured by glucometry. The homeostasis model assessment of IR (HOMA-IR) was used to assess maternal IR, calculated using the formula $\text{HOMA-IR} = \text{insulin } (\mu\text{IU/mL}) \times \text{glucose (mmol/L)} / 22.5$, as originally described by (Matthews et al., 1985; Rabasa-Lhoret et al., 2003). Although HOMA-IR was initially developed for use in humans and a species-specific correction factor for adult ruminants has not been established, this index has been widely used in bovine studies as a relative indicator of insulin sensitivity for within-population comparisons. In adult cattle, normal fasting blood glucose concentrations are typically maintained at ≤ 3.9 mM (Bell and Bauman, 1997). In the present study, cows classified in the maternal-high-IR group exhibited mean glucose concentrations of 4.76 mM during the dry period, exceeding the established upper physiological range and indicating impaired glucose regulation. For each cow, HOMA-IR values were calculated at each sampling point, and the mean HOMA-IR across wk 1 to 7 of the dry period was used for group classification. Based on these averaged values, cows were retrospectively stratified into 2 groups: maternal low IR ($n = 22$; mean HOMA-IR: 4.21 ± 0.16) and maternal high IR ($n = 22$; mean HOMA-IR: 6.07 ± 0.12). These ranges are consistent with previously reported values for dry dairy cows during late gestation (De Koster and Opsomer, 2013; Hasegawa et al., 2019). Variation in HOMA-IR within cows across sampling weeks was moderate and followed consistent temporal patterns, supporting the use of averaged values to represent overall maternal IR status during dry period.

The cows were housed in environmentally controlled semi-open barns designed to maintain an optimal temperature range, thereby minimizing cold stress. The TMR was provided ad libitum twice daily at 0800 and 1700 h. No clinical diseases or veterinary interventions occurred during the study period, ensuring that the measured metabolic outcomes were not confounded by health complications.

Calf Rearing and Health Scoring

Calves were separated from dams within 1 h postpartum and housed in individual pens within a well-ventilated, enclosed barn with sidewalls. The barn was designed to minimize drafts while allowing adequate air exchange. Ambient environment conditions were monitored throughout the experimental period. The mean ambient temperature during the study was $12.4^\circ\text{C} \pm 5.2^\circ\text{C}$ (range: 2.8°C to 22.1°C), and the mean temperature-humidity index was 52.3 ± 6.1 . Although the average temperature was above the lower critical temperature reported for young dairy calves (9°C), ambient temperature intermittently fell below this threshold. Calves were exposed to temperatures below 9°C (Silva and Bittar, 2019) for approximately 2.8% of the observation period. To mitigate cold exposure, pens were bedded with dry straw for insulation, cleaned daily with fresh bedding, and spaced 0.5 m apart to prevent physical contact between calves. Throughout the study, calves were observed daily for visible signs of cold stress, including shivering, piloerection, abnormal posture, hunching, reduced activity, and decreased milk intake; these signs were not observed. Each calf received pooled colostrum from maternal cows (6 L per feeding, twice daily) within 12 h of birth. Colostrum chemical composition was 3.49% fat, 3.75% protein, 5.67% lactose, 11.88% TS, and 11.8 mg/dL MUN. Following colostrum feeding, calves were fed 6 L of pasteurized whole milk twice daily until 30 d of age. Whole milk composition averaged 3.19% fat, 3.35% protein, 5.27% lactose, 12.88% TS, 12.1 mg/dL MUN, and 4.9×10^4 cells/mL SCC. Whole milk samples were collected weekly and composited for nutrient analysis. All calves consumed the full allotted milk volume at each feeding; therefore, milk refusals were not observed. Calves had ad libitum access to fresh water and starter feed. The starter containing 88.6% DM, and chemical composition (DM basis) included 25.7% CP, 3.27% fat, 10.4% ADF, 20.5% NDF, 28.5% starch, 6.04% ash, 1.76 Mcal/kg NEL (Supplemental Table S2, see Notes). Starter intake was recorded daily and did not differ between maternal-low-IR and maternal-high-IR groups (Supplemental Table S3, see Notes; $P \geq 0.23$). All calves were fed and managed by trained research staff to ensure con-

sistent feeding practices, intake recording, and animal handling across treatment groups. Rectal temperature was measured at 0, 2, 4, 6, 8, 16, and 24 h postpartum. Body weight, withers height, and body length were recorded weekly during the rearing period, with withers height measured according to the standardized protocol by Lesmeister and Heinrichs (2005).

Calves were monitored 3 times per week for health parameters, including fecal consistency, cough, nasal discharge, and respiration rate. Fecal consistency was scored on a 0-to-3 scale, with 0 indicating normal consistency and scores of 2 or higher classified as diarrhea. Coughing was assessed on a 0-to-3 scale, where 0 indicated no cough and higher scores indicated increasing severity. Nasal discharge was rated on a 0-to-3 scale, with 0 representing no discharge and higher scores reflecting more pronounced discharge. Respiration rate was measured by counting abdominal movements per minute in resting calves, as described by Spain and Spiers (1996). This method was used to evaluate respiratory health and detect signs of distress. Health scoring followed the protocols established by the University of Wisconsin–Madison Veterinary Medicine guidelines (Edmonson et al., 1989), ensuring consistency and accuracy in evaluating calf health.

Blood Sampling and Analysis for Dams

Blood samples were collected from dry cows 1 h before the morning feeding via coccygeal venipuncture using 15-mL vacuum tubes at wk 1, 3, 5, and 7 after drying off. Samples were centrifuged at $3,000 \times g$ for 15 min at 4°C, and serum was stored at –20°C for later analyses. Serum insulin concentrations were measured at using ELISA kits (#MM-5100301, Meimian; #BOFI00145, Genie) according to the manufacturers' protocols (Malven et al., 1987). Glucose concentrations were determined at corresponding time points using test strips (#07124112, Roche Diagnostics).

Glucose Tolerance Testing

For the glucose tolerance testing, 4-wk-old calves were fasted for 12 h to ensure consistent baseline glucose levels, with ad libitum access to water throughout the fasting period to prevent dehydration. After fasting, 16-gauge $\times$ 14-cm catheters coated with lithium heparin were aseptically inserted into the jugular vein of each calf. Catheter placement was performed under strict aseptic conditions to prevent contamination or infection and to ensure accurate blood sampling. A glucose solution (250 mg/kg BW) was prepared by dissolving the appropriate amount of glucose in 20 mL of saline. This glucose solution was infused intravenously at 0700 h to

standardize the test time across all subjects. The glucose infusion protocol followed established methods (Bach et al., 2013; Pantophlet et al., 2016; Yohe et al., 2021), ensuring consistency across different experimental setups. After infusion, blood samples were collected at specific intervals: 0, 15, 30, 45, 60, and 90 min after the glucose infusion. Blood glucose concentrations were measured at each time point using glucometer strips (Roche Diagnostics), allowing real-time, reliable glucose assessment. The use of glucometer strips ensured rapid measurement and minimized handling time, which is critical for reducing potential variations in glucose readings due to sample degradation or external factors. To enhance test reliability, calves were closely monitored for any signs of distress during the procedure.

Backfat Thickness Measurement

Backfat thickness was measured using a B-mode ultrasound (ImaGo S, IMV Imaging) equipped with a 5.0 to 7.5 MHz transducer, set to a depth range of 80 to 100 mm. The sites to be scanned were first cleared of debris and then gelled for optimal image acquisition. The ultrasound images were interpreted on-site, with a measurement accuracy of 0.1 mm (Schröder and Staufienbiel, 2006).

Tissue Sample Collection for Calves

Calves were killed for tissue collection either within 24 h after birth ($n = 20$) or at 1 mo of age (30.33 ± 0.76 d of age, $n = 24$). All procedures were conducted in the morning following a 6 h fast in the on-farm dissection room. Calves were killed by captive-bolt stunning followed by exsanguination via severance of the jugular vein. Immediately after the calves were killed, perirenal BAT was collected. During dissection, the renal pelvis and surrounding connective tissues were carefully removed to ensure consistent and complete harvesting of the perirenal fat depot across calves. The isolated BAT depot was weighed using an electronic balance. Samples were then placed into sterile specimen bags and either rapidly frozen in liquid nitrogen and stored at –80°C for subsequent molecular analyses or fixed in 4% paraformaldehyde (G1101; Servicebio) for histological evaluation.

Hematoxylin and Eosin and Histologic Immunostaining

The BAT samples were fixed in 4% paraformaldehyde at 4°C for 24 h, paraffin-embedded, and sectioned at 5- μ m thickness. Hematoxylin and eosin staining was performed according to previously described protocols (Liu et al., 2020; Oeckl et al., 2022). For each calf, 3 par-

affin sections separated by at least 50 μm were analyzed. From each section, 3 non-overlapping microscopic fields were randomly selected and imaged at 200 $\times$ magnification using a light microscope, resulting in a total of 9 images per calf. Adipocyte area and cell number were quantified using the Deconvolution2 plugin in ImageJ software (version 1.53c, National Institutes of Health). Individual adipocytes were identified based on clear cellular boundaries, and adipocyte number was expressed as the number of cells per image field.

For immunofluorescence, endogenous peroxidase activity was quenched with 3% H_2O_2 , followed by blocking with BSA. Primary antibodies against UCP1 (1:200; #A5857, ABclonal), DLK1 (1:200; #A2715, ABclonal), and DRP1 (1:200; #A21968, ABclonal) were applied overnight at 4°C. After washing, sections were incubated with Cy3-conjugated secondary antibodies (1:1,000; #AS007, ABclonal). Mitochondria were stained with MitoSPY (1:1,000; #C1048, Beyotime), and nuclei were counterstained with 4',6-diamidino-2-phenylindole (DAPI; 5 $\mu\text{g}/\text{mL}$; #RM02978, ABclonal). Images were captured using a Leica DMi8 microscope, and fluorescence intensity was normalized to background levels to ensure accurate measurement of signal intensity. This method enables visualization and quantification of protein expression and mitochondrial dynamics in BAT sections.

Brown Adipogenic Differentiation In Vitro

To determine whether maternal IR alters offspring BAT development at the tissue level, mRNA and protein expression of key thermogenic markers were directly measured in BAT collected from calves at birth and 1 mo of age. In addition, an in vitro stromal vascular fractions (SVF) culture model was employed as a complementary approach to access the intrinsic differentiation capacity and thermogenic potential of brown adipocyte precursor cells under controlled hyperglycemic and hyperinsulinemic conditions. This in vitro system was to mechanistically validate the in vivo findings while minimizing systemic confounding factors.

Specifically, SVF were isolated from neonatal calf BAT by collagenase I digestion (#BS163, Biosharp) at 37°C for 1 h in a shaking incubator (Han et al., 2015). Cells were cultured in low-glucose Dulbecco's Modified Eagle Medium (DMEM; Gibco) supplemented with 10% fetal bovine serum (FBS; #U11-059A, Yobibio). Digested fractions were filtered through 100- μm cell strainers, and the supernatant was removed by centrifugation at 1,000 $\times g$ for 5 min at 4°C. Isolated SVF were pooled and cultured in DMEM supplemented with 10% FBS, 100 U/mL penicillin, and 1,000 U/mL streptomycin (#15240062, Gibco) at 37°C in a 5% CO_2 incubator (Wci-260, Wiggins). Culture medium was re-

placed every 2 d. For brown adipocyte differentiation, the isolated SVF were seeded in 6-well plates. Upon reaching confluence, SVF were induced to differentiate into brown adipocytes in DMEM supplemented with 125 μM indomethacin (#I811784, Macklin), 0.5 mM 3-isobutyl-1-methylxanthine (#I811775, Macklin), 1 μM dexamethasone (#BL171B, Biosharp, China), 5 $\mu\text{g}/\text{mL}$ insulin (#I828365, Macklin), and 1 μM 3,3',5-triiodo-L-thyronine (T3; #T834800, Macklin) for 4 d. Cells were then further induced in DMEM with 5 $\mu\text{g}/\text{mL}$ insulin and 1 μM T3 for 3 d to promote adipocyte maturation (Chen et al., 2021; Tan and Jiang, 2024). During the differentiation, SVF were also treated for adipogenic induction: Confluent SVF were treated with control (5 mM glucose and 5 ng/mL insulin), moderate glucose (25 mM glucose and 5 ng/mL insulin), high glucose (50 mM glucose and 5 ng/mL insulin), moderate insulin (5 mM glucose and 25 ng/mL insulin), high insulin (5 mM glucose and 50 ng/mL insulin), moderate IR (25 mM glucose and 25 $\mu\text{g}/\text{mL}$ insulin), and high IR (50 mM glucose and 50 $\mu\text{g}/\text{mL}$ insulin).

Oil Red O Staining

Differentiated adipocytes were fixed (10% formalin, 15 min, room temperature) for lipid accumulation visualization. Lipid was assessed by Oil Red O (#AC11564, Acme) staining. Cells were stained with a working solution of Oil Red O in 60% isopropanol for 10 min to visualize intracellular lipid droplets. For quantification, Oil Red O was dissolved in isopropanol and measured at 520 nm using microplate reader (Synergy HTX microplate reader, BioTek Instruments; Kwan et al., 2017). Additional method details were presented in Supplemental File S1 (see Notes).

Statistical Analysis

Statistical analyses were performed using R 4.4.0 (R Core Team, 2024) with *lme4* package (v.1.1-35), and data visualization was conducted using GraphPad Prism 8.0 (GraphPad Inc., San Diego, CA). A power test was performed using the *simr* package to determine the least replicates cattle required to detect a biological difference in maternal blood IR. The test indicated that the minimum sample size of $n > 4$ per group could gain over 80% power at $\alpha 0.05$ using fitted model in current study. Maternal cows and their offspring calves served as the experimental units. Data normality was assessed using the Shapiro–Wilk test in the *stats* package, and homogeneity of variances was evaluated using Levene's test. Variables that did not meet the assumptions of normality or equal variance were log-transformed before analysis. Linear mixed effects models with repeated

Table 1. Dry cow intake, BW, BCS, backfat thickness (BFT), and blood concentrations of insulin, HOMA-IR, and glucose in dry cows included in this study

Item	Treatment ¹			P-value		
	Maternal-low-IR	Maternal-high-IR	SEM	Treatment	Time	Treatment × time
Dry cow parameters						
DMI (kg/d)	12.7	13.1	0.71	0.57	0.42	0.84
BW (kg)	711	728	9.37	0.15	<0.01	0.75
BCS	3.2	3.4	0.06	0.20	<0.01	0.98
BFT (mm)	30.1	31.17	0.21	0.82	<0.01	0.96
Blood parameters						
Glucose (mM)	3.88	4.76	0.09	<0.01	<0.01	<0.01
Insulin (μIU)	27.39	23.67	0.36	<0.01	<0.01	<0.01
HOMA-IR ²	4.05	5.81	0.15	<0.01	<0.01	<0.01

¹Maternal-low-IR: dry cows with normal insulin resistance. Maternal-high-IR: dry cows with high insulin resistance.

²HOMA-IR: Homeostasis model assessment of insulin resistance. $\text{HOMA-IR} = [\text{fasting insulin } (\mu\text{IU/mL})] \times [\text{fasting glucose } (\text{mmol/L})] / 22.5$.

measures were applied to analyze response variables such as maternal blood parameters, BCS, BW, healthy scores, calf body condition, and body temperature. The models included fixed effects of treatment (maternal IR), time, and their interaction, with cow specified as a random effect. The covariance structure for repeated measures was specified by nesting time within cow using the model syntax (Time | Cow) in the *lme4* package. For other variables without repeated measures, such as BAT mass, gene expression, protein expression, and in vitro experimental outcomes, were analyzed using mixed model, including fixed effect of treatment and random effect of cow or plate. Results are presented as mean ± SEM. Significance was set at $P < 0.05$.

RESULTS

Maternal Performance and Blood Insulin, HOMA-IR, and Glucose

Table 1 presents dry cow DMI, BW, BCS, BFT, and blood concentrations of insulin, HOMA-IR, and glucose in dry cows (Table 1). No significant differences were observed in DMI ($P = 0.57$), BW ($P = 0.15$), BCS ($P = 0.20$), or BFT ($P = 0.82$) between maternal-low-IR and maternal-high-IR groups. However, serum glucose and insulin levels were significantly higher in the maternal-high-IR group compared with the maternal-low-IR group ($P < 0.01$). Similarly, HOMA-IR was significantly elevated in the maternal-high-IR group compared with the maternal-low-IR group ($P < 0.01$).

Calf Performance

No significant differences were observed in BW (Figure 1A; $P = 0.13$), withers height (Figure 1B; $P = 0.11$), body

length (Figure 1C; $P = 0.57$), or rectal body temperature (Figure 1D; $P > 0.17$) between calves from the maternal low IR and maternal high IR at 1 mo old. Meanwhile, monitoring of growth parameters, weekly BW change ($P \geq 0.65$), withers height change ($P \geq 0.21$), and body length change ($P \geq 0.10$), and rectal body temperature ($P > 0.17$) over 1 mo revealed no significant differences between the maternal-low-IR and maternal-high-IR groups (Figure 1E–H, Supplemental Table S4, see Notes).

Postnatal Glucose Homeostasis and Health Scores

Glucose tolerance tests conducted in 1-mo-old calves showed impaired glucose clearance as evidenced by significantly elevated blood glucose levels and a greater area under the curve in the maternal-high-IR group (Figure 1H; $P < 0.01$). Calf health was monitored throughout the preweaning period, and no significant differences were observed in overall health scores or growth rates between groups. However, calves born to maternal-high-IR group exhibited numerically elevated clinical morbidity indicators (Table 2), including higher fecal scores ($P = 0.05$), cough scores ($P = 0.07$), and incidence of diarrhea tendency ($P = 0.09$).

Postnatal BAT Morphometrics and Function

Calves born to high-IR dams exhibited significantly reduced perirenal BAT mass (Figure 2A, maternal high IR vs. maternal low IR; 120.8 ± 4.5 vs. 180.0 ± 7.2 g, newborn; 138.8 ± 11.0 vs. 194.2 ± 4.3 g, 1 mo of age; $P < 0.01$) and increased brown adipocyte size (Figure 2B, maternal high IR vs. maternal low IR, 562.5 ± 26.28 μm^2 vs. 466.7 ± 12.61 μm^2 , newborn; 680.8 ± 22.92 μm^2 vs. 424.2 ± 13.70 μm^2 , 1 mo of age; $P < 0.01$) in calves. Moreover, mRNA expression of thermogenic genes,

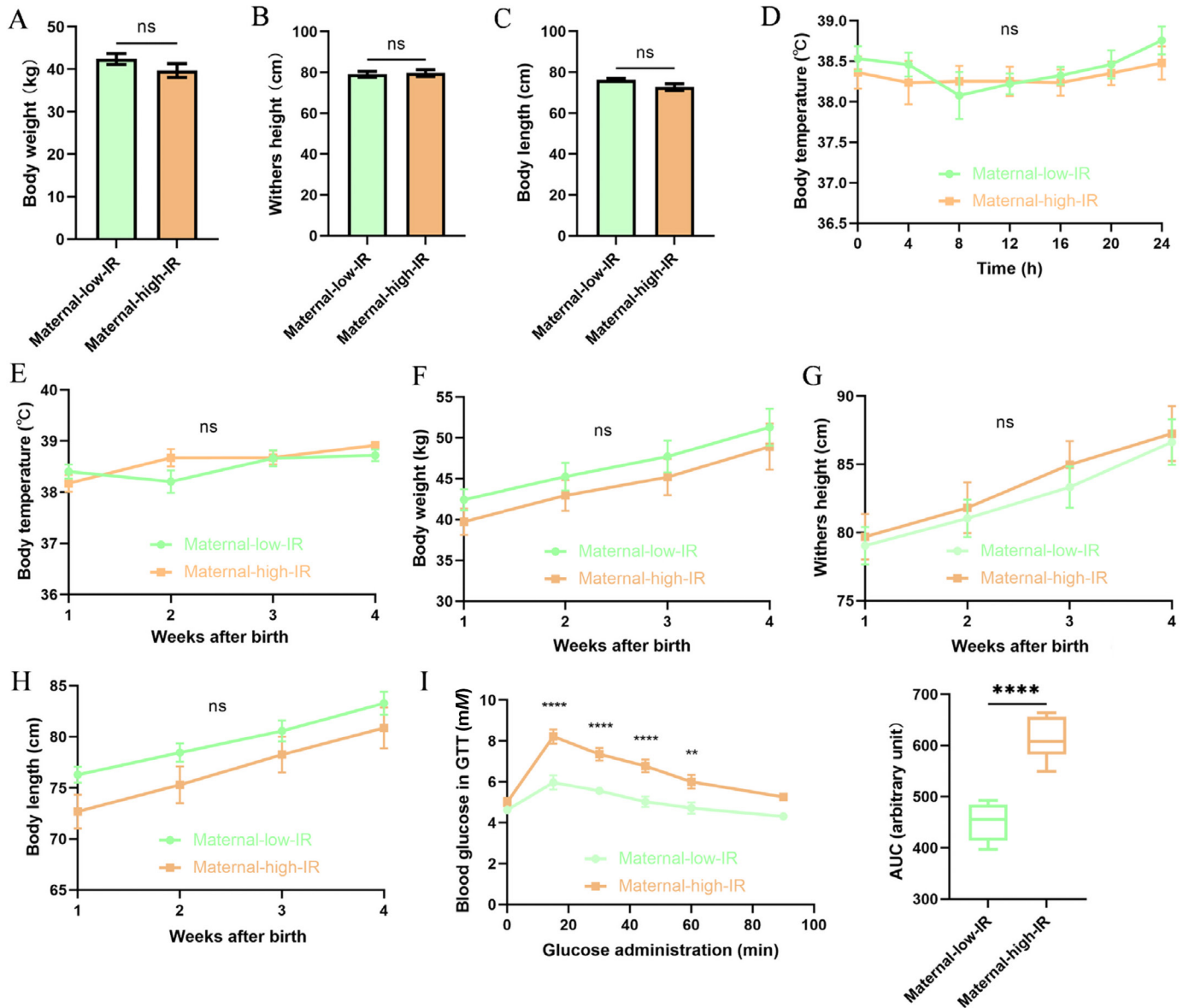


Figure 1. Effects of maternal insulin resistance (IR) levels on calf growth and glucose metabolism during the first month postpartum. Body weight (A), withers height (B), and (C) body length of calves born to dry cows with normal insulin resistance (maternal low IR; $n = 10$) or high insulin resistance (maternal high IR; $n = 12$) at 1 mo of age. (D) Rectal body temperature measured at multiple time points within the first 24 h after birth. (E–H) Longitudinal changes in calf physiological and growth parameters during the first month postpartum, including rectal temperature (E), BW (F), withers height (G), and body length (H). (I) Blood glucose concentrations measured during a glucose tolerance test (GTT) at multiple time points, along with area under the curve (AUC) analysis. (Upper and lower edges of the boxes represent the 75th and 25th percentiles, respectively; the center line represents the mean; and whiskers represent SEM.) Data are presented as mean $\pm$ SEM. Significant difference is indicated by asterisks (** $P < 0.01$, **** $P < 0.0001$; ns = not significant).

including uncoupling protein 1 (UCP1), and lipogenesis genes including cAMP response element-binding protein 4 (CREBP4) and carnitine palmitoyl transferase 1B (CPT1B) was significantly downregulated, accompanied by reduced protein levels of UCP1 ($P < 0.01$) and PGC-1 α ($P = 0.02$; Figures 2C–D). Furthermore, mitochondrial density in BAT was also markedly reduced in the maternal-high-IR group (Figure 2E; $P < 0.01$), in

alignment with the observed decreases in UCP1 protein content. These impairments persisted at 1 mo of age.

BAT Glucose Uptake, Adipogenic Development, and Mitochondrial Dynamics

Glucose uptake in BAT is primarily mediated by insulin receptor (INSR) and glucose transporter 4 (GLUT4).

Table 2. The healthy scores in calves born to maternal-low-IR and maternal-high-IR groups included in this study

Item	Treatment ¹			P-value		
	Maternal low IR	Maternal high IR	SEM	Treatment	Time	Treatment × time
Fecal score ² (0–3)	0.89	1.48	0.14	0.07	0.03	0.84
Cough score ² (0–3)	0.84	1.42	0.17	0.04	0.06	0.81
Diarrhea ³ (%)	14.4	30.9	3.92	0.09	—	—
Respiration rate (breaths/min)	37.8	41.3	1.34	0.21	0.01	0.56

¹Maternal low IR: dry cows with normal insulin resistance. Maternal high IR: dry cows with high insulin resistance.

²Adapted from University of Wisconsin–Madison School of Veterinary Medicine (2011).

³Diarrhea was fecal score ≥ 2 .

Calves from the maternal-high-IR group showed significantly reduced mRNA and protein (Figure 3; $P < 0.01$) levels of both INSR and GLUT4 at birth and 1 mo of age.

Notably, expression of DLK1 was markedly upregulated at both the mRNA and protein levels in BAT from maternal-high-IR calves (Figures 3A–C; $P < 0.01$), with immunofluorescence confirming increased DLK1 staining at birth and 1 mo (Figure 3D; $P < 0.01$). Additionally, protein levels of mitochondrial fission regulators dynamin-related protein 1 (DRP1) and fission 1 protein (FIS1) were significantly decreased at both time points (Figure 3; $P < 0.01$).

Effects of Hyperglycemia-Hyperinsulinemia on BAT Development In Vitro

Oil Red O staining revealed significantly reduced lipid droplet formation in the SVF derived preadipocytes under combined glucose-insulin treatments, indicating suppressed adipogenic differentiation (Figure 4A, B). Gene expression analysis demonstrated downregulation of critical thermogenic *UCP1* (Figure 4C; $P < 0.01$), upregulation of adipogenetic precursor inhibitor *DLK1* (Figure 4C; $P < 0.01$), and decreased expression of mitochondrial fission marker *DRP1* (Figure 4C; $P < 0.01$), particularly under high glucose-insulin conditions. Western blot analysis confirmed reductions in INSR, UCP1, and mitochondrial fission DRP1 protein levels in cells exposed to combined hyperglycemic and hyperinsulinemic treatments (Figure 4D; $P < 0.01$).

DISCUSSION

Balanced energy supply, particularly maternal glucose levels, is critical for fetal development, neonatal survival, and subsequent growth performance in calves (Bach, 2012; Radunz et al., 2012). Although DMI was recorded, precise energy balance calculations were not performed for dams, and that the underlying cause for divergent IR status in late gestation warrants further study.

However, elevated maternal glucose levels and IR during late gestation are common metabolic disturbances in cows that not only impair maternal productivity but also pose significant health risks to the offspring (De Koster and Opsomer, 2013; Sundrum, 2015). Maternal hyperglycemia and IR can adversely affect placental function, suppress IGF-1 signaling, and ultimately disrupt fetal growth (Hay, 2006; Hasegawa et al., 2019; Dos Santos et al., 2024). Brown adipose tissue plays a pivotal role in glucose metabolism, especially in neonates (Bienboire-Frosini et al., 2023). Although extensive research in humans and rodents has explored the effects of maternal glucose dysregulation on fetal BAT development, the effect on BAT development and metabolic programming in cattle remains poorly understood. In this study, we demonstrate that maternal IR did not significantly affect calf birth weight or early postnatal growth. However, it markedly impaired neonatal glucose metabolism, BAT development, mitochondrial function, and elevated DLK1 signaling. Furthermore, in vitro experiments with high glucose and IR-like conditions recapitulated these in vivo defects, showing reduced glucose metabolism, impaired adipogenic differentiation, including impaired glucose metabolism, reduced preadipocyte differentiation, mitochondrial dysfunction, and increased DLK1 signaling activation. These findings establish a mechanistic link between maternal hyperglycemia in late gestation and impaired BAT development and function in calves.

During late gestation, maternal glucose requirements increase markedly due to rapid fetal growth, contributing to the high incidence of IR and fat cow syndrome in dairy herds (Brockman, 1978; Catalano, 2010). In our study, maternal glucose levels in the high glucose and IR group averaged 4.76 mM, exceeding the upper threshold of the normal physiological range in adult cattle (≤ 3.9 mM; Bell and Bauman, 1997). Previous studies in humans and rodents have shown that maternal IR during pregnancy can significantly influence fetal development and neonatal energy homeostasis (Levin, 2006; Henriksson et al., 2015). In particular, maternal hyperglycemia and hyper-

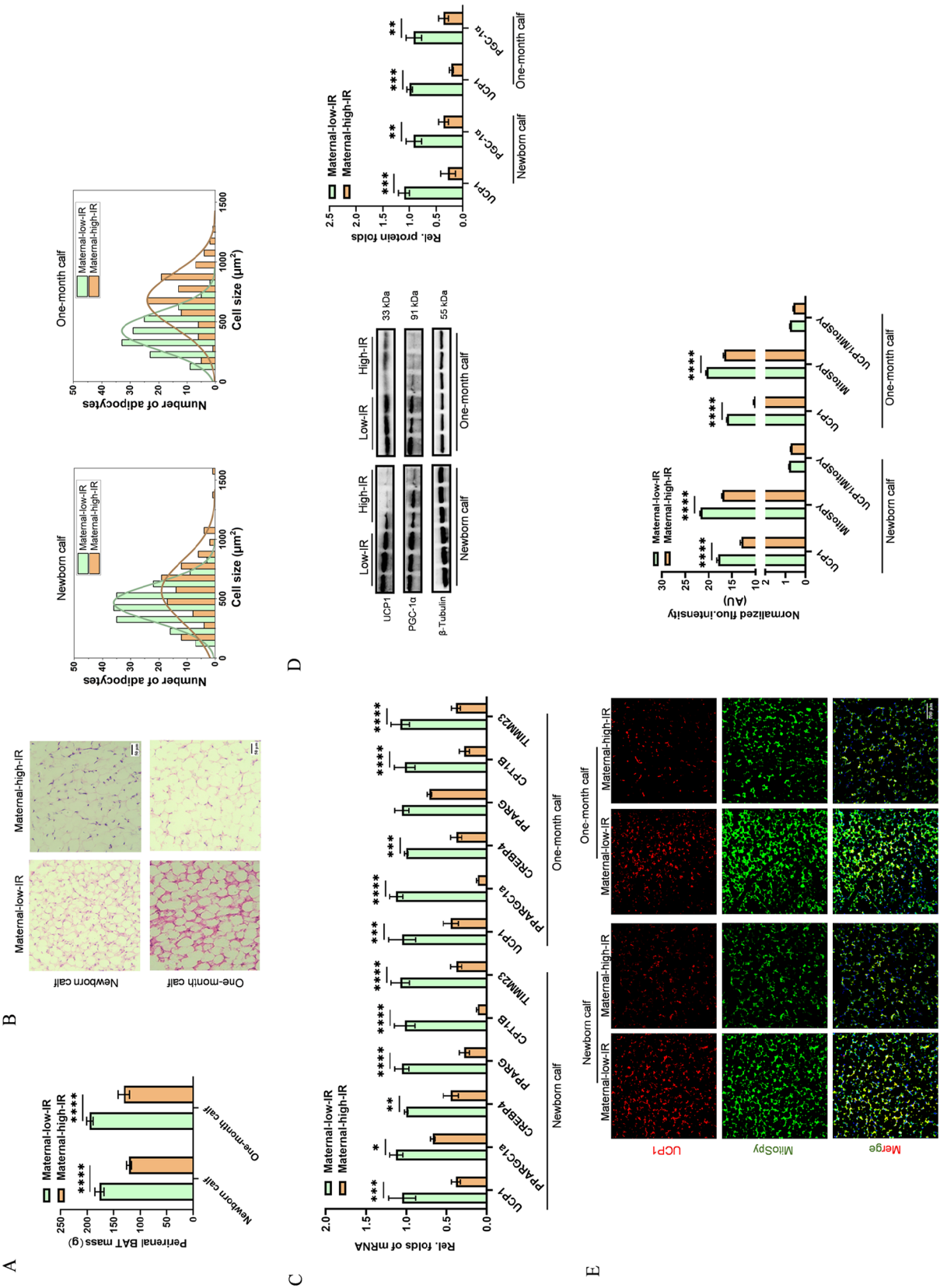
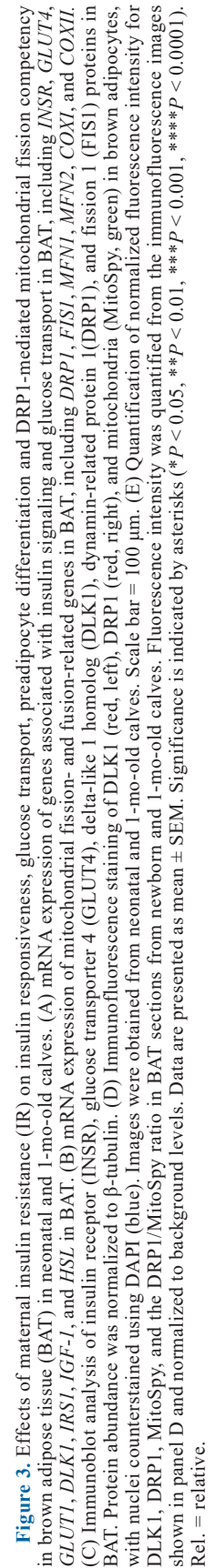


Figure 2. Effects of maternal insulin resistance (IR) on brown adipose tissue (BAT) development and thermogenic function in newborn and 1-mo-old calves. (A) Perirenal BAT mass. (B) Hematoxylin and eosin staining of perirenal BAT. Scale bar = 50 μ m. (C) Quantitative real-time PCR analysis of key thermogenic and adipogenic gene expression in BAT, including *UCP1*, *PPARGC1 α* , *CREBP4*, *PPARG*, *CPT1B*, and *TIMM23*. (D) Western blot analysis of UCP1 and PGC-1 α protein expression in BAT, normalized to β -tubulin. (E) Immunofluorescence staining of BAT sections for UCP1 (red) and the mitochondrial marker (MitoSpy, green), with nuclei counterstained with DAPI (blue). Fluorescence (fluor.) intensity was quantified using ImageJ. Scale bar = 100 μ m. Data are presented as mean $\pm$ SEM. Significance is indicated with asterisks (* P < 0.05, ** P < 0.01, *** P < 0.001, **** P < 0.0001). AU = arbitrary units; Rel. = relative.



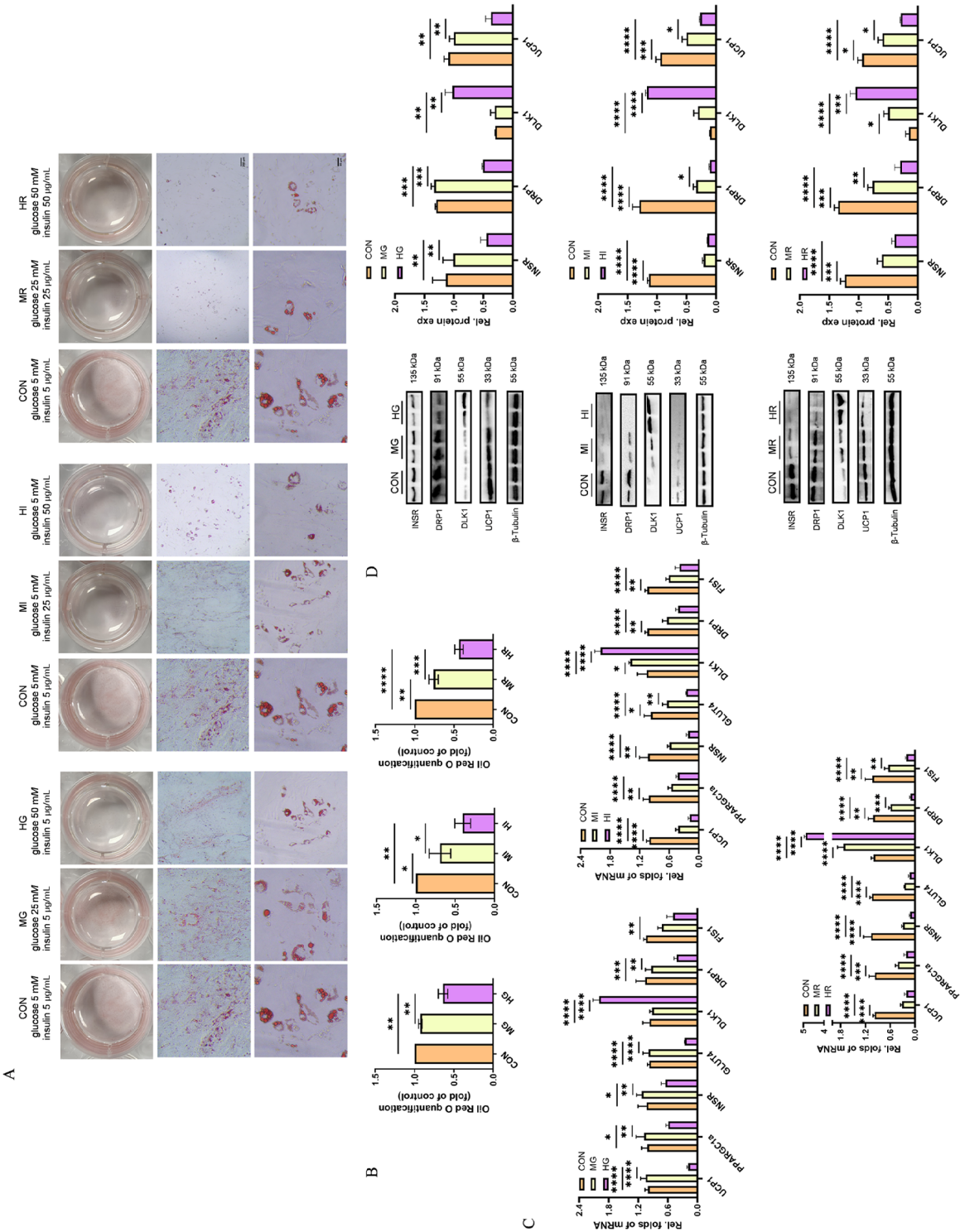


Figure 4. Glucose and insulin supplementation inhibit brown adipogenesis, DRP1-mediated mitochondrial fission, and preadipocyte differentiation in vitro (A) Oil Red O staining of differentiated brown adipocytes treated with various concentrations of glucose or insulin during adipogenic induction, with PBS as the control. Representative images were captured at 10× (scale bar = 50 µm) and 40× (scale bar = 200 µm) magnification. (B) Quantification of lipid accumulation in differentiated adipocytes under control, low (0.5 mM), and high (2 mM) glucose and insulin treatments (n = 3). (C) mRNA expression of genes associated with thermogenesis, mitochondrial fission and preadipocyte differentiation, including *UCP1*, *PPARGC1α*, *INSR*, *GLUT4*, *DRP1*, *FIS1*, and *DLK1*. (D) Immunoblot analysis of UCP1, DRP1, INSR, and DLK1 proteins in differentiated brown adipocytes, normalized to β-tubulin. Data are presented as mean ± SEM. Significance is indicated by asterisks (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$). Rel. = relative; exp = expression. CON = control, 5 mM glucose + 5 µg/mL insulin; MG = 25 mM glucose + 5 µg/mL insulin; HG = 50 mM glucose + 5 µg/mL insulin; MI = 5 mM glucose + 25 µg/mL insulin; HI = 5 mM glucose + 50 µg/mL insulin; MR = 25 mM glucose + 25 µg/mL insulin; HR = 50 mM glucose + 50 µg/mL insulin.

insulinemia are associated with excessive fetal growth and increased risk of macrosomia (Schwartz et al., 1994). In contrast, our results indicate that maternal glucose levels of 4.77 mM had limited effects on early postnatal growth metrics in calves (≤ 1 mo of age), including BW, body length, withers height, body temperature, and health scores. This suggests that although the maternal glucose elevation was insufficient to cause significant physical growth difference, it was sufficient to impair normal BAT development and function in calves.

Fetal BAT undergoes rapid proliferation during late gestation and can comprise up to 5% of neonatal BW (Symonds et al., 2015). In cattle, BAT is primarily located in the neck, perirenal, and pericardial regions, with perirenal BAT accounting for approximately 50% of total BAT mass (Smith et al., 2004). Our study focused on perirenal BAT. Previous studies suggest that maternal glucose hyperglycemia can suppress fetal BAT development by impairing glucose uptake and utilization (Moley et al., 1998; Yu et al., 2019). In agreement, we observed significant reduction in perirenal BAT mass in calves from hyperglycemic dams, indicating compromised thermogenic capacity and an increased risk of energy imbalance in calves.

Brown adipose tissue is essential for nonshivering thermogenesis in newborns, contributing to body temperature regulation immediately after birth. In adults, although BAT activity declines, it continues to contribute to energy partitioning and regulation of glucose metabolism (Carpentier et al., 2018; Bienboire-Frosini et al., 2023). Brown adipose tissue dysfunction is associated with IR, and increased risk of metabolic disorders (Gonzalez-Franquesa and Patti, 2017; Desjardins and Steinberg, 2018). In this study, the perirenal BAT from calves born to high-IR dams exhibited significantly reduced expression of insulin responsiveness markers, such as *INSR* and *GLUT4*, suggesting compromised insulin sensitivity. In parallel, thermogenesis-related genes, including *UCP1* and *PGC1 α* , were also downregulated in BAT, indicating compromised BAT functionality and increased vulnerability to metabolic stressors in early life.

Given that BAT thermogenesis depends on mitochondrial oxidative activity, particularly at the mitochondrial membrane, alterations in mitochondrial dynamics likely contribute to functional impairment (Cedikova et al., 2016; Chouchani et al., 2017). In calves from high-IR dams, we observed significant downregulation of mitochondrial fission-related genes, such as *DRP1* and *FIS1*, indicating suppressed mitochondrial remodeling capacity and BAT thermogenesis. Notable, fusion-related genes, including *MFN1* and *MFN2*, however, remained unchanged, indicating an imbalance in fission-fusion processes. This imbalance may lead to reduced mitochondrial efficiency and ATP production, contributing to

glucose accumulation, exacerbated IR, and further BAT dysfunction. These results underscore the importance of maintaining mitochondrial dynamics in maintaining BAT thermogenic function and systemic metabolic homeostasis (Babbar and Sheikh, 2013; Gonzalez-Franquesa and Patti, 2017; Moorthy et al., 2024).

Brown adipose tissue originates from preadipocyte differentiation, a process tightly regulated by molecular signals such as *DLK1*, a conserved inhibitor of adipogenesis (Boeuf et al., 2001; Carobbio et al., 2013; Hudak and Sul, 2013). *DLK1* is a preadipocyte marker known to suppress brown adipocyte differentiation, *UCP1* expression, and adipogenic gene expression (Huff et al., 2005; Charalambous et al., 2014). In our study, maternal-high-IR calves showed elevated *DLK1* expression in BAT, consistent with impaired adipogenic differentiation and BAT function. These findings suggest that dysregulated *DLK1* signaling is a key mediator of maternal hyperglycemia-induced BAT dysfunction.

To further investigate the mechanisms by which hyperglycemia and IR impair BAT development, we conducted in vitro experiments with high glucose and IR-like conditions. These experiments confirmed that both high glucose and insulin, independently and synergistically, inhibited brown adipocyte differentiation, as evidenced by reduced lipid droplet formation and downregulation of key thermogenic genes (*UCP1*, *PPARGC1 α* , and *CREBP4*) and proteins (*UCP1*, *INSR*). Mitochondrial fission was impaired (reduced *DRP1* expression), whereas *DLK1* signaling was upregulated, recapitulating the in vivo findings. These results validate that hyperglycemia and hyperinsulinemia negatively affect brown adipogenesis and mitochondrial function, highlighting a potential programming mechanism by which maternal metabolic state affects calves BAT development.

Although our study underscores *DLK1* signaling as a central pathway in mediating these effects, additional regulatory mechanisms may also be involved. For example, *NOTCH* signaling, which interacts with *DLK1*, has been shown to inhibit adipogenesis (Nueda et al., 2018; Rodríguez-Cano et al., 2020). Furthermore, because *DLK1* can induce the synthesis and secretion of insulin through the ERK and AKT pathways, it may have distinct roles in modulating BAT differentiation and *UCP1* expression (Sul et al., 2010; Li et al., 2025). Future studies should explore these complementary pathways to better define the molecular network underlying maternal metabolic influence on calves BAT development.

Although calf starter intake did not differ between treatment groups, individual variations in feeding behavior or nutrient utilization may not be fully captured and therefore cannot be entirely excluded as contributors to postnatal metabolic outcomes. In addition, although no overt behavioral signs of cold stress were observed,

we cannot exclude the possibility that intermittent exposure to temperatures below the lower critical threshold induced physiological cold stress that was not apparent behaviorally and may have influenced BAT-related outcomes. Classification of cows into maternal-low-IR and maternal-high-IR groups was based on circulating glucose, insulin and HOMA-IR during the dry period; however, the physiological mechanisms leading certain cows to develop greater IR remain unclear. Potential contributors may include differences in metabolic adaptability, prior lactation history, body fat distribution, hepatic insulin sensitivity, or inflammatory status, which were not directly assessed in the present study. Therefore, although associations between maternal IR and impaired offspring BAT development were identified, the causal factors underlying the emergence of IR in late gestation cows warrant further investigation.

CONCLUSIONS

Maternal IR and hyperglycemia during late gestation significantly impair BAT development and function in neonatal calves. Although maternal IR had minimal effects on birth weight and early postnatal growth, it resulted in impaired glucose metabolism, compromised BAT development, and mitochondrial dysfunction in offspring. Calves from high-IR dams showed reduced perirenal BAT mass, increased brown adipocyte size, down-regulated thermogenic gene expression, and decreased mitochondrial density. These effects persisted at 1 mo of age, indicating long-term impairment of thermogenic function. In vitro experiments supported these findings, demonstrating that hyperinsulinemia disrupted brown adipocyte differentiation and mitochondrial dynamics. Elevated *DLK1* expression in BAT from maternal-high-IR calves further indicated impaired adipogenic differentiation. These results underscore the critical role of maternal metabolic health in offspring metabolic programming and highlight potential mechanisms, including *DLK1* signaling and mitochondrial dynamics, that mediate BAT dysfunction. Further research into additional regulatory pathways and potential interventions to mitigate these effects is essential for improving neonatal health and metabolic outcomes.

NOTES

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Nonstandard abbreviations used: AU = arbitrary units; AUC = area under the curve; BAT = brown adipose tissue; BFT = backfat thickness; CON = control, 5 mM glucose + 5 µg/mL insulin; DAPI = 4',6-diamidino-2-phenylindole; DMEM = Dulbecco's Modified Eagle Medium; exp = expression; FFA = free fatty acids; GTT = glucose tolerance test; HG = 50 mM glucose + 5 µg/mL insulin; HI = 5 mM glucose + 50 µg/mL insulin; HOMA-IR = homeostasis model assessment of IR; HR = 50 mM glucose + 50 µg/mL insulin; IR = insulin resistance; MG = 25 mM glucose + 5 µg/mL insulin; MI = 5 mM glucose + 25 µg/mL insulin; MR = 25 mM glucose + 25 µg/mL insulin; ns = not significant; Rel. = relative; SVF = stromal vascular fractions; T3 = 3,3',5-triiodo-L-thyronine; THI = temperature-humidity index.

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ORCIDS

- Guiling Ma,  <https://orcid.org/0009-0008-9786-5460>
- Morteza Hosseini Ghaffari,  <https://orcid.org/0000-0002-5811-3492>
- Yanting Chen  <https://orcid.org/0000-0001-7319-7893>